3 Methodology [2 pages]

The project employed the CRISP-DM framework to ensure a structured and systematic approach towards completion of the ML model. The steps in CRISP-DM include Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment. Figure 3.1 summarises the steps in CRISP-DM.

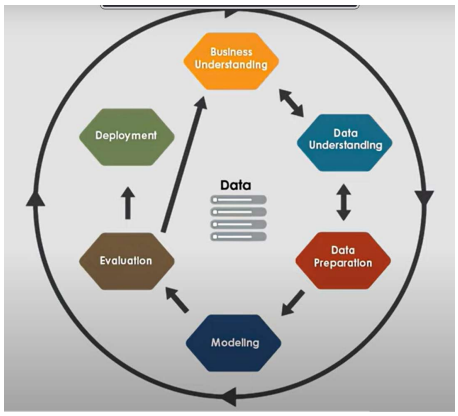


Figure 3.1: CRISP-DM Process Flow

During the business understanding phase, a literature review was conducted into Machine Learning in Crime and GDP. The literature review found that many have attempted to implement Machine Learning in this domain. Hence, the problem was zoomed into a sub-set of the problem that could be solved within a short project duration by 2 team members i.e. crime and GDP in Malaysia.

During the Data Understanding phase, OpenDOSM datasets were downloaded and interpreted. The primary dataset used was crime_district.csv. This dataset is supplemented by the secondary dataset gdp_district_real_supply.csv (Annual Real GDP by District & Economic Sector, 2025). Initial analysis based on the description found on OpenDOSM and based on domain knowledge study found that the data would not be easily compatible for analysis without proper data massaging procedures.

During this phase as well, we also used a Power BI dashboard to understand the trends and patterns in the data. Figure 3.2 shows the Power BI dashboard.

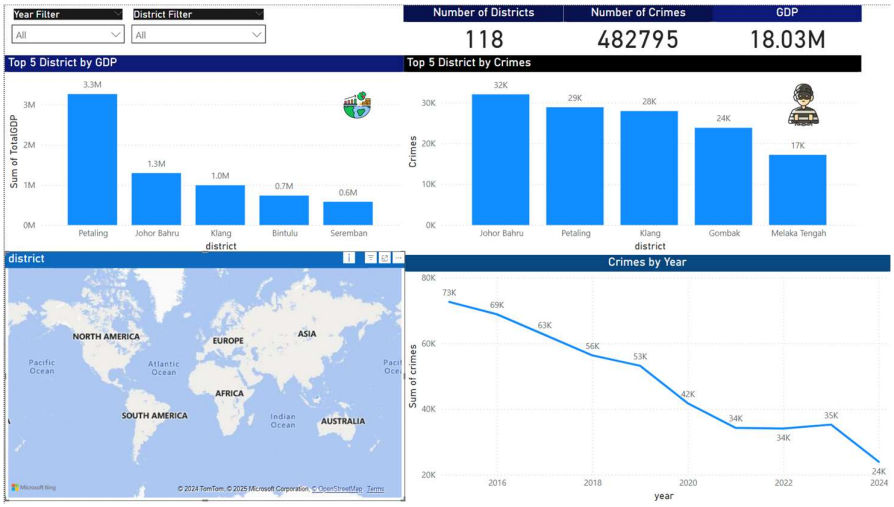


Figure 3.2: Power BI dashboard

3.1 Data Preparation

During this step, the data will be massaged and transformed into a format that would be compatible with each other for analysis. The crime and GDP data are in their raw format as per below:

crime_district.csv			
district	year	state	crimes
Alor Gajah	2015	Melaka	424
Alor Gajah	2016	Melaka	436
Alor Gajah	2017	Melaka	376

gdp_district_real_supply.csv					
series	state	district	date	sector	value
abs	Johor	Batu Pahat	1/1/2015	p0	11605.5
abs	Johor	Batu Pahat	1/1/2015	p1	2177.257
abs	Johor	Batu Pahat	1/1/2015	p2	13.0113

Table 3.1: Sample Data before Data Preparation

Seeing as this data is not formatted in a district-year-variable format, it needs to be massaged and transformed (on both sides) into this format before they can be merged. This requires data deduplication to remove crime categories from the crime data and data massaging to pivot rows to columns for the GDP data on the right.

The data will then be merged and transformed using some of the sample code attached in Appendix. The result of the transformation (pivoting, duplicate removal, missing value treatment, merging etc.). Once the datasets are merged, some final preparation is needed before it is ready for modelling. In the state it is in Table 3.2, the data would only be useful to predict the crimes of a district in the same year the GDP is made available. In terms of practical applications, this is not very helpful. Hence, the feature columns of the dataset need to be lagged by a year (shifted one row up) for each district. The code to do this is also included in the Appendix and the resulting dataset sample is shown in Table 3.2.

year	district	crimes	gdp_purchasers_l1	gdp_agriculture_l1	gdp_manufacturing_l1	gdp_services_l1
2015	Alor Gajah	424	1477	726.8712	10.7435	729.7303
2015	Bachok	364	1395	434.7939	194.1093	719.3754
2015	Baling	376	600	265.7816	35.6816	286.8733

Table 3.2: Sample Data after Data Preparation

3.2 Supervised Learning (Regression)

The target variable is crime while the features are gdp_purchasers_l1, gdp_agriculture_l1, gdp_manufacturing_l1, and gdp_services_l1 which represent the respective GDP values of that district and in the year before. District and year are excluded from the machine learning model because there are insufficient examples of each district, to allow the model to learn (district has a very high cardinality and cannot be encoded). The year is not deemed a relevant variable in this problem area.

First, correlation analysis will be done to find the variables that have a correlation to the target variable. Based on the correlation analysis, decisions can be made on the features that can be selected and excluded. Then, the regression models will be trained in separate experiments involving 5 models of different varieties. The best model that has a good trade off of performance without any concessions will be selected as the best model.

3.3 Unsupervised Learning (Clustering)

An unsupervised learning approach will also be used to identify clusters of crime and GDP. The same dataset prepared for regression will be used for clustering. All variables (both crime and GDP variables) will be used as features. The algorithms to be used are KMeans clustering, Agglomerative clustering, and DBScan.

The data will first need to be scaled for all features to be on the same range, and to ensure that they are all well-distributed throughout the scale. The code to perform scaling is included in the Appendix. Then, the Elbow method is used to determine the suitable number of clusters. For KMeans clustering, the Elbow method for inertia will be used. For Agglomerative clustering, the dendogram will be used. Finally, for DBScan, the Elbow Method with Epsilon will be plotted. Then, the silhouette score will be evaluated and the best performing model with a meaningful number of clusters will be selected as the model to proceed with which can be interpreted for insights. Insights such as different blends and patterns of crime and GDP that are associated with one another could be determined, for example – which are the crime rate thresholds for varying levels of crime (very high, high, low, very low), and what are the associated GDP of these high crime rate district.

4 Results

4.1 Regression

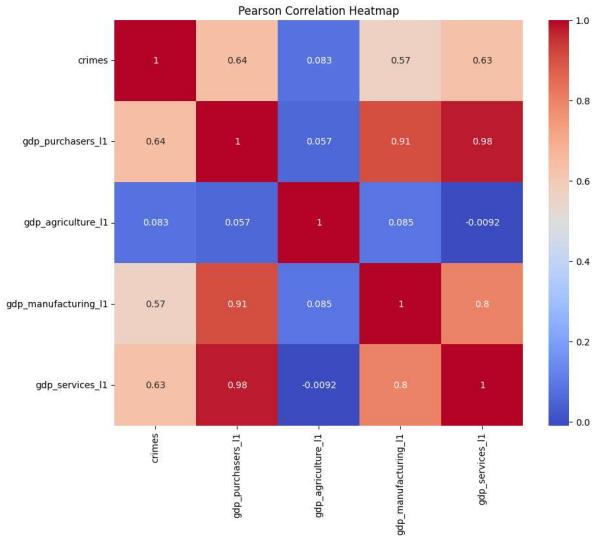


Figure 4.1: Correlation of crimes to gdp features

The correlation analysis in Figure 4.2 shows that there is a high correlation in the GDP values to crime for all GDP sectors except for gdp_agriculture. This intuitively makes sense because one would think that agriculture of a district doing good or bad does not have any bearing on crimes.

There is also a high correlation between the features. We found that this might cause a multicollinearity problem in the model but as we will see later, the model is performing well even with this multicollinearity (with an R2 of more than 80%) and hence, it is safe to assume that this is not a problem.

Based on these correlation analysis insights, we made the decision to implement 2 versions of models i.e. one version with all features, and one version with all features excluding agriculture. The table summarises the performance of the two versions of the model, for all 5 model types that were tried.

Version	Model	MSE	R2	CV Score	Comments
All Features	KNN Regressor	0.183590	0.811343	0.743294	Best non-overfitted model
	Random Forest Regressor	0.130680	0.865713	0.732096	Best R2, but overfitted
	SVR	0.151774	0.844037	0.664049	Overfitted model
	ElasticNet	0.844439	0.132254	0.118033	Worst model in every way
Without agriculture	KNN Regressor	0.241139	0.752206	0.718310	Best non-overfitted model
	Random Forest Regressor	0.233957	0.759586	0.688725	Best R2, but overfitted
	SVR	0.307226	0.684294	0.639601	Overfitted model
	ElasticNet	0.844439	0.132254	0.118033	Worst model in every way

Table 4.1: Summary of Regression Results

As shown in Table 4.4, the results were compared, and it was found that the performance of the model (measured by R2) does not significantly drop by removing agriculture beyond a confidence level of 95%. Considering the correlation analysis and the model results, we decided to remove agriculture from the final model. This is because despite the model performing slightly worse (but not significantly worse), the agriculture feature might be introducing noise to the model and not helping it learn the crimes.

KNN Regressor is the superior model for both versions hence, it is selected as the model to be used for future predictions. This model is ideal not just because it performs well but also because the size of the data is not that big i.e. only about 1000 rows. KNN Regressor struggles with data of large size and can be computationally expensive to run on large data. This is not a problem for our use case because the data is relatively small i.e. only 1000 rows, and this data will not grow much beyond this point because it's size only increases by about 150 news for every year (there are only about 150 districts in Malaysia). This could be looked at again in the future when scaling to other countries is considered but for now, it is deemed fine to opt for KNN regression.

4.2 Clustering

First, the Elbow Method based on inertia measurement resulting from different algorithms yielded the following results:

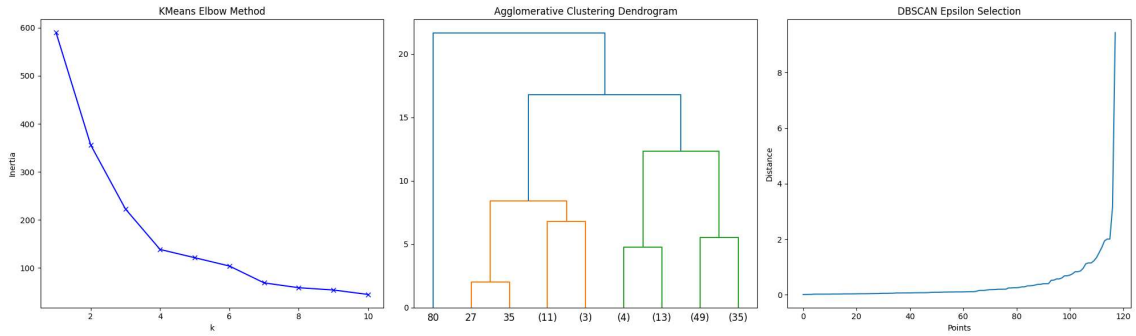


Figure 4.2: Ideal Number of Clusters for Unsupervised Learning

For KMeans clustering, the ideal cluster number is selected as 4, while for Agglomerative clustering, the ideal clusters found are 4 clusters as well. For DBScan, the ideal epsilon is about 2. After setting this hyperparameter for the models, the performance in terms of silhouette score was measured. It was found that Agglomerative clustering performed the best in terms of the clustering models.

Model	Silhouette Score	Comments
DBScan	0.779413	An epsilon of 2 means clusters of noise and not noise. Not meaningful.
Agglomerative	0.546694	The next best model. With 4 clusters, it is meaningful.
KMeans	0.539633	The worst model.

Table 4.2: Clustering Results Summary

As such, the agglomerative clustering model was then used to assign the clusters for each district and year and then the clusters were interpreted for descriptive statistics and similarities that could explain the clusters.

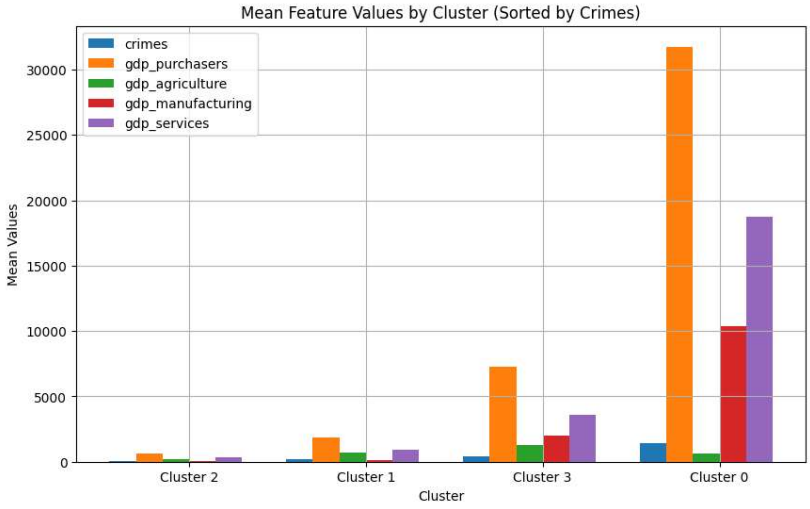


Figure 4.3: Clusters Interpretation for Agglomerative Clustering Model

Based on the insights, it was found that an average crimes of 68.82, 170.61, 413.27, and 1396.02 are very low, low, high, and very high number of crimes for a district. These clusters of crime also show a trend of increasing gdp_purchasers which indicates that high crime clusters also have high gdp_purchasers. Hence, it might be better for the government of Malaysia to place crime fighting efforts in places with gdp_purchasers.